The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 July 1995

INCH-POUND

MIL-S-19500/595B
27 April 1995
SUPERSEDING
MIL-S-19500/595A
1 September 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT,
TRANSISTOR, P-CHANNEL, SILICON,
TYPES 2N7236, 2N7237, 2N7236U, AND 2N7237U
JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a P-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500, and two levels of product assurance for each unencapsulated device type die, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}).

1.2 Physical dimensions. See figure 1 (TO-254AA), figure 2 for surface mount devices, and figure 3 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings (Unless otherwise specified, T_C = +25°C).

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = -1.0 mA dc	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C	I_S	I_{DM} 3/	T_{op} and T_{STG}	$R_{\theta JC}$ max
	V dc	W	W	V dc	A dc	A dc	A dc	A(pk)	°C	°C/W
2N7236	-100	125	4.0	±20	-18	-11	-18	-72	-55	1.0
2N7237	-200	125	4.0	±20	-11	-7	-11	-44	to	1.0
2N7236U	-100	75	4.0	±20	-14	-9	-14	-56		1.67
2N7237U	-200	75	4.0	±20	-8	-5	-8	-32	+150	1.67

See footnotes on next page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, Ohio 45444-5765 by using the Standardization Document Approvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA
FSC 5961

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Type	I_{AR} 2/	E_{AS}	E_{AR}	$r_{DS(on)}$ max 1/ 4/ V_{GS} = -10 V dc $I_D = I_{D2}$	
				T_J = +25°C	T_J = +150°C
	A	mj	mj	ohm	ohm
2N7236	-18	500	12.5	0.20	0.400
2N7237	-11	500	12.5	0.51	1.122
2N7236U	-18	500	7.5	0.20	0.400
2N7237U	-11	500	7.5	0.51	1.122

1/ Derate linearly 1.0 W/°C (for non-"U" suffix devices) or 0.6 W/°C (for "U" suffix devices) for $T_C > +25°C$; $P_T = \frac{T_{J(max)} - T_C}{R_{\theta JX}}$

2/ $I_D = \sqrt{\frac{T_{J(max)} - T_C}{(R_{\theta JX}) \times (R_{DS(on)} \text{ at } T_{J(max)})}}$

3/ $I_{DM} = 4I_{D1}$; I_{D1} as calculated in footnote 2/.

4/ Pulsed (see 4.5.1).

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = -1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = -0.25 mA dc		Max I_{DSS1} V_{GS} = 0	Max $r_{DS(on)}$ 2/ V_{GS} = -10 V dc $I_D = I_{D2}$
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25°C
	V dc	V dc		μA dc	Ohms
		Min	Max		
2N7236	-100	-2.0	-4.0	25	0.20
2N7237	-200	-2.0	-4.0	25	0.51
2N7236U	-100	-2.0	-4.0	25	0.20
2N7237U	-200	-2.0	-4.0	25	0.51

1/ Unless otherwise specified, electrical characteristics for "U" suffix devices are identical to the corresponding non "U" devices.
2/ Pulsed (see 4.5.1).

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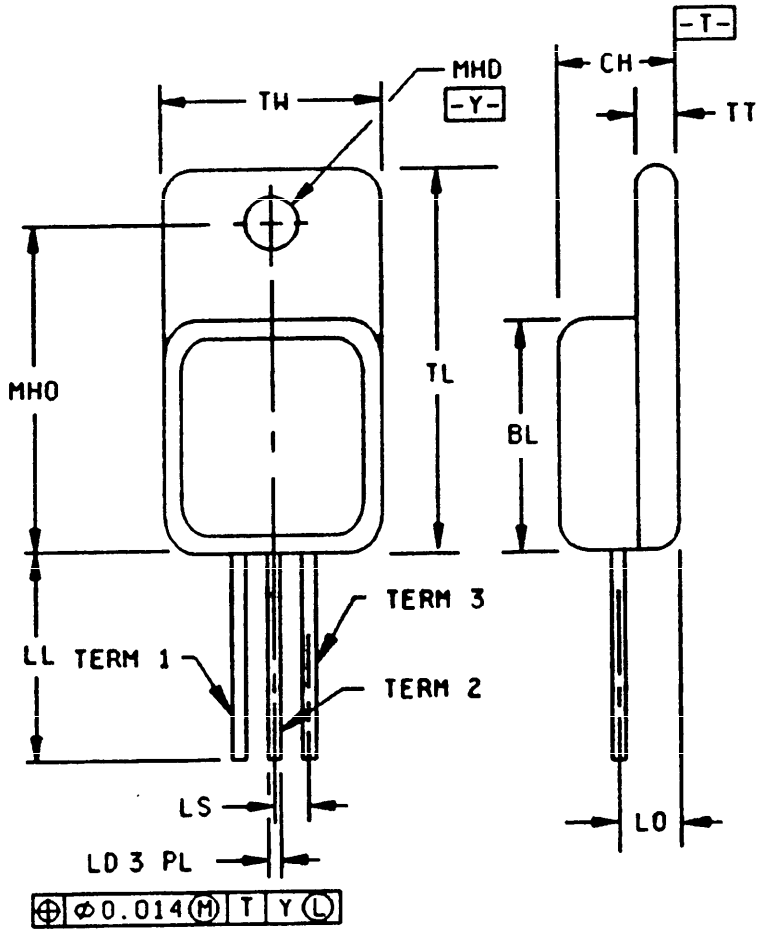


FIGURE 1. Physical dimensions for TO-254AA.

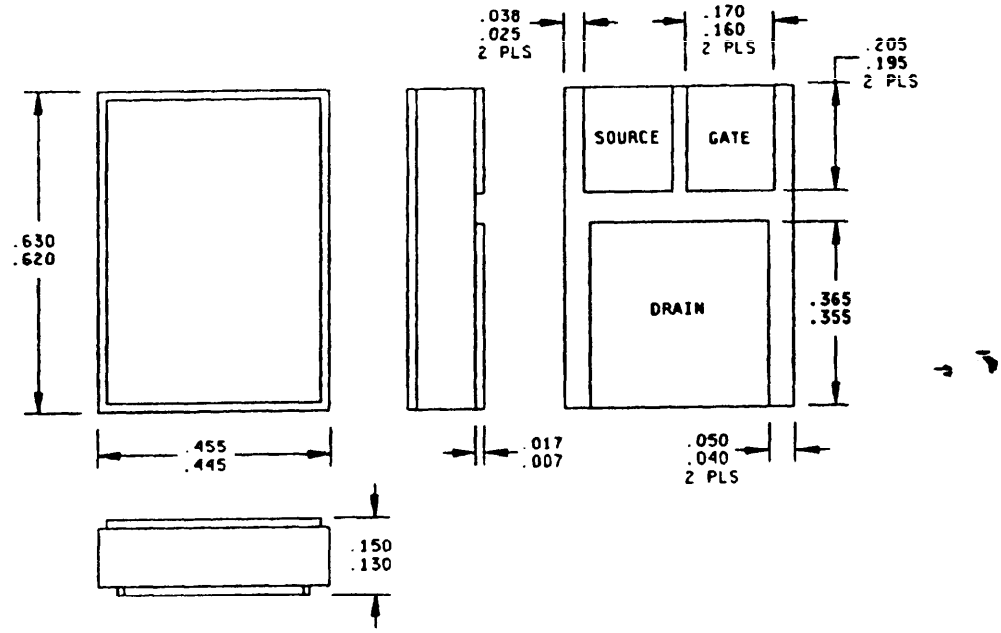
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Ltr	Dimensions				
	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CH	.249	.260	6.32	6.60	
TT	.040	.050	1.02	1.27	
LO	.150 BSC		3.81 BSC		
TL	.790	.800	20.07	20.32	
BL	.535	.545	13.59	13.84	
LS	.150 BSC		3.81 BSC		
TW	.535	.545	13.59	13.84	
LL	.530	.550	13.46	13.97	
MHO	.665	.685	16.89	17.40	
MHD	.139	.149	3.53	3.78	
LD	.035	.045	0.89	1.14	
Term 1	Drain				3
Term 2	Source				3
Term 3	Gate				3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All terminals are isolated from the case.

FIGURE 1. Physical dimensions for TO-254AA - Continued.

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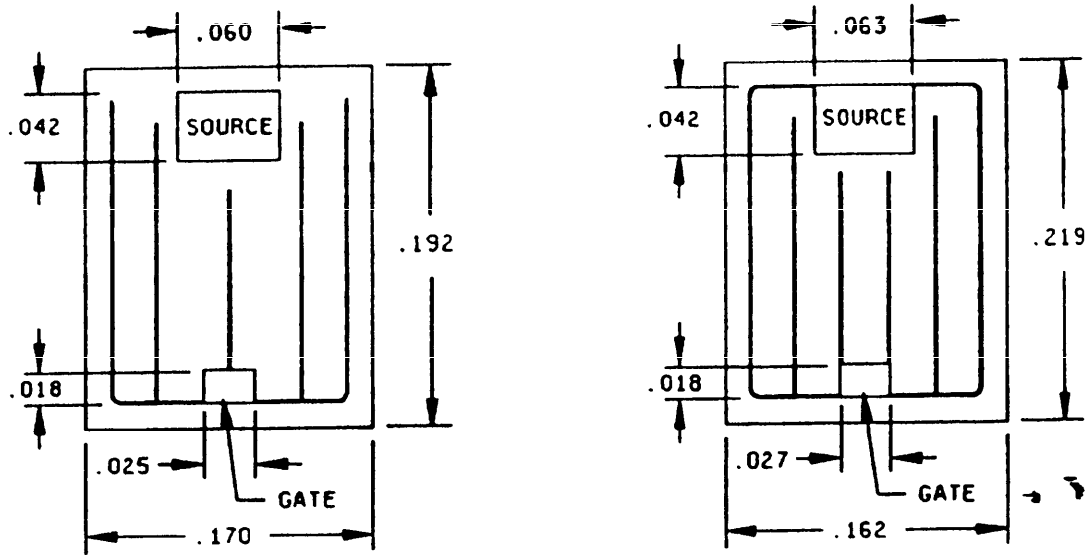


Inches	mm	Inches	mm
.007	0.18	.170	4.32
.017	0.43	.195	4.95
.025	0.64	.205	5.21
.038	0.97	.355	9.02
.040	1.02	.365	9.27
.050	1.27	.445	11.30
.130	3.30	.455	11.56
.150	3.81	.620	15.75
.160	4.06	.630	16.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).

FIGURE 2. Dimensions and configuration of surface mount package outline.

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2N7236 2N7237

A version

Inches	mm
.018	0.46
.025	0.64
.027	0.69
.042	1.07
.060	1.52
.063	1.60
.162	4.11
.170	4.32
.192	4.88
.219	5.56

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±0.005 inch (0.13 mm).
4. Physical characteristics of the die thickness = .0187 inch (0.47 mm).
5. Back metal: Cr - Ni - Ag.
6. Top metal: Al.
7. Back contact: Drain.

FIGURE 3. Physical dimensions JANHC and JANKC.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 40 (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as specified herein.

JANH	High reliability product assurance level for unencapsulated devices.
JANK	Space reliability product assurance level for unencapsulated devices.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, and 3 herein. Methods used for electrical isolation of the terminal feed through shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.3.1 Lead formation material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead formation material or finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table II of MIL-S-19500 and 100 percent DC testing in accordance with group A, subgroup 2 herein.

3.3.2 Internal construction. Multiple chip construction shall not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

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3.4.1 Marking of JANHC and JANKC die. For JANHC and JANKC die container, the following marking shall be used (example):

JANHCAM2N7236
↑↑↑↑
|||└ RHA level (see MIL-S-19500). 1/
||└ Source of manufacturer (see figure 3).
|└ Unencapsulated.
└ Product assurance level 2/.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. R ≤ 100 k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be in accordance with table II herein.

4.2.2 JANHC and JANKC devices. Qualification shall be in accordance with appendix H of MIL-S-19500.

1/ RHA (included only for radiation hardened devices) (see MIL-S-19500, appendix G).

2/ Two levels of product assurance levels are provided for unencapsulated devices, H and K (see MIL-S-19500.)

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4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9 1/	I_{GSS1}, I_{DSS1} Subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$
12	MIL-STD-750, method 1042, condition A	MIL-STD-750, method 1042, condition A T_A = +175°C and t = 48 hours
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent initial value. $\Delta V_{GS(th)1}$ = ±20 percent initial value.

1/ Shall be performed anytime before screen 10.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500, (appendix H), as a minimum die shall be 100 percent probed in accordance with group A, subgroup 2, except test current shall not exceed 20 A.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table I, subgroup 2 herein.)

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, strength test may be performed after C6.
B4	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.
B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at I_D = -1 mA dc. Read and record I_{DSS} (pre and post) in accordance with table I, subgroup 2 herein. $V_{BR(DSS)}$ delta cannot exceed 10 percent.
B5	1042	Accelerated steady-state gate stress; test condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
B6		See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds, t = 10 s (not applicable for surface mount devices).
C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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Dimensions are in inches.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = 1.0°C/W for TO-254AA case style devices and 1.67°C/W for surface mount devices. The following parameter measurements shall apply:

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	3.3 A (2 A for surface mount devices).
c.	t_H heating time	Steady state (see MIL-STD-750, method 3161).
d.	V_H drain-source heating voltage	25 V (20 V for surface mount devices).
e.	t_{MD} measurement time delay	30 to 60 µs.
f.	t_{SW} sample window time	10 µs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed the group A, subgroup 2 limit or figure 4 thermal impedance curve) for $Z_{\theta JC}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line monitor. The following parameter measurements shall apply:

a.	I_M measuring current	10 mA.
b.	I_H drain heating current	3.3 A (2 A for surface mount devices).
c.	t_H heating time	100 ms (50 ms for surface mount devices).
d.	V_H drain-source heating voltage	25 V (20 V for surface mount devices).
e.	t_{MD} measurement time delay	30 to 60 µs.
f.	t_{SW} sample window time	10 µs maximum.

4.5.4 Single pulsed unclamped inductive switching.

a.	Peak current, I_D	I_{AR}(max).
b.	Peak gate voltage, V_{GS}	-10 V.
c.	Gate to source resistor, R_{GS}	25 ≤ R_g ≤ 200 ohms.
d.	Initial case temperature	+25°C, +10°C, -5°C.
e.	Inductance, L	$\left\lfloor 2E_{AS}/(I_{D1})^2 \right\rfloor \left\lfloor \frac{(V_{BR} - V_{DD})}{V_{BR}} \right\rfloor$ mH minimum.
f.	Number of pulses to be applied	1 pulse minimum.
g.	Supply voltage (V_{DD})	-25 V for 2N7236, -50 V for 2N7237.

4.5.5 Gate stress test. V_{GS} = ±30 V minimum; t = 250 µs minimum.

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Electrical parameters not

TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 3/	3161		$Z_{\Theta JC}$			°C/W
Breakdown voltage, drain to source	3407	I_D = -1.0 mA dc, Bias condition C, V_{GS} = 0 V dc	$V_{(BR)DSS}$			
2N7236				-100		V dc
2N7237				-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -.25 mA dc	$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate reverse current	3411	V_{GS} = +20 V dc and -20 V dc, Bias condition C, V_{DS} = 0	I_{GSS1}		±100	nA dc
Drain current	3413	V_{DS} = 80 percent of rated V_{DS}, Bias condition C, V_{GS} = 0	I_{DSS1}		-25	µA dc
Static drain to source "on"-state resistance	3421	V_{GS} = -10 V dc, condition A, Pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7236					0.20	ohm
2N7237					0.51	ohm
Static drain to source "on"-state resistance	3421	V_{GS} = -10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7236					0.22	ohm
2N7237					0.52	ohm
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), I_D = I_{D1} (see 1.3)	V_{SD}			
2N7236					-4.2	V
2N7237					-4.6	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate reverse current	3411	V_{GS} = +20 V dc and -20 V dc, Bias condition C, V_{DS} = 0	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V dc				
		V_{DS} = 100 percent rated V_{DS} (see 1.3)	I_{DSS2}		-1.0	mA dc
		V_{DS} = 80 percent rated V_{DS} (see 1.3)	I_{DSS3}		-0.25	mA dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = -.25 mA dc	$V_{GS(th)2}$	-1.0		V dc
Static drain to source "on"-state resistance	3421	V_{GS} = -10 V dc, Pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)3}$			
2N7236					0.34	ohm
2N7237					1.10	ohm
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -.25 mA dc	$V_{GS(th)3}$		-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = Rated I_{D2} (see 1.3), V_{GS} = -10 V dc, Gate drive impedance = 9.1 Ω, V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$		35	ns
Rise time			t_r		85	ns
Turn-off delay time			$t_{d(off)}$		85	ns
Fall time			t_f		65	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area test	3474	See figure 5 (derate by a factor of 1.67 for "U" suffix devices), V_{DS} = 80 percent of rated V_{DS}, V_{DS} = -200 V maximum, t_p = 10 ms				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			nC
2N7236				31	60	
2N7237				28	60	
Charge gate to source			Q_{gs}			nC
2N7236				3.7	13	
2N7237				3.0	15	
Charge gate to drain			Q_{gd}			nC
2N7236				7.0	35.2	
2N7237				4.5	38	
Reverse recovery time	3473	$V_{DD} \leq 30$ V, $d_i/d_t \leq -100$ A/µs, $I_D = I_{D1}$ (see 1.3)	t_{rr}			ns
2N7236					280	
2N7237					440	

1/ For sampling plan, see MIL-S-19500.
2/ Unless otherwise indicated, inspections, conditions, and limits for "U" suffix devices are identical to corresponding non-suffix devices.
3/ This test is required for the following endpoint measurement only (not intended for screen 13): JANS, group B, subgroups 3 and 4; JANTX and JANTXV, group B, subgroups 2 and 3; group C, subgroup 6, and group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels except JANHC and JANKC) for qualification only.

Inspection 1/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = o
Temperature cycling	1051	500 cycles, test condition G	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 2/			45 devices c = o
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = see 4.5.2	
Subgroup 5			
Not applicable			
Subgroup 6			5 devices c = 0
Repetitive avalanche energy	3469	Peak current $I_{AR} = I_D$; Peak gate voltage V_{GS} = -10 V; Gate to source resistor, R_{GS} 25 ≤ R_{GS} ≤ 200 Ω Temperature = T_J = +150°C +0°C, -10°C Inductance = $\left[\left[2E_{AR}/(I_{D1}{}^2)\right]\right]\left[\left[\frac{(V_{BR}-V_{DD})}{V_{BR}}\right]\right]$ mH min Number of pulses to be applied = 3.6×10^8; supply voltage (V_{DD}) = -50 V for 2N7237 and 2N7237U, (V_{DD}) = -25 V for 2N7236 and 2N7236U, time in avalanche = 2 µs minimum, 20 µs maximum; f = 500 Hz minimum.	
Electrical measurements		See table I, group A, subgroup 2	

1/ JANHC and JANKC device are qualified with appendix H of MIL-S-19500.
2/ A separate sample for each test may be pulled.

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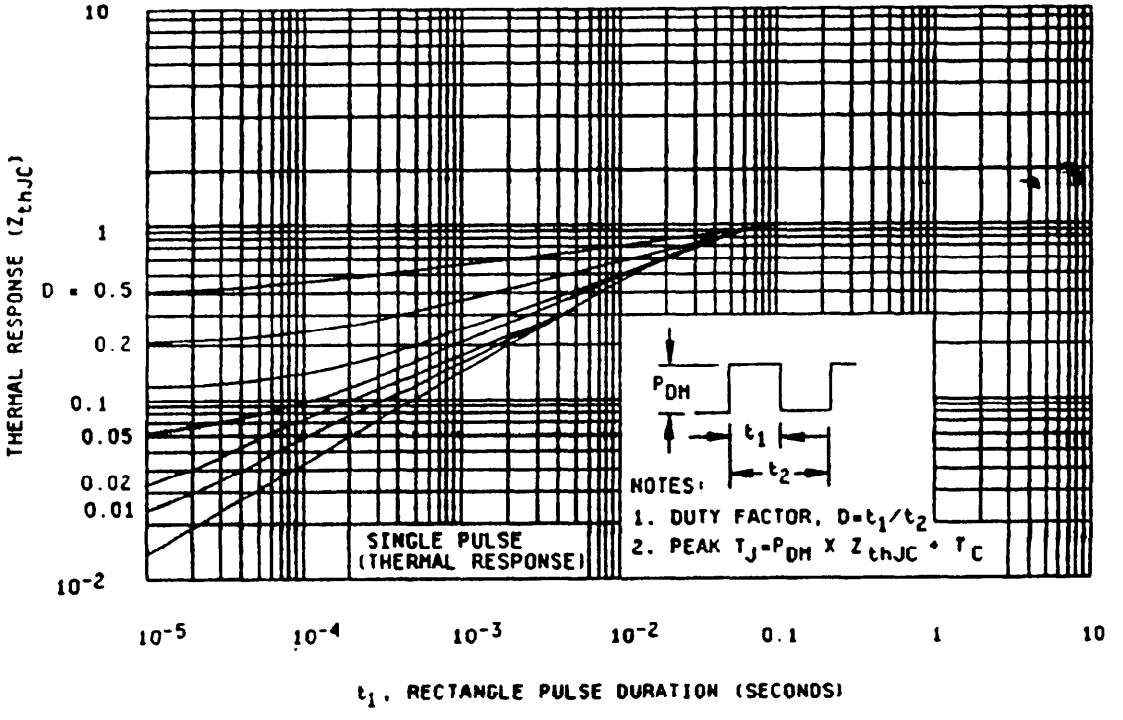
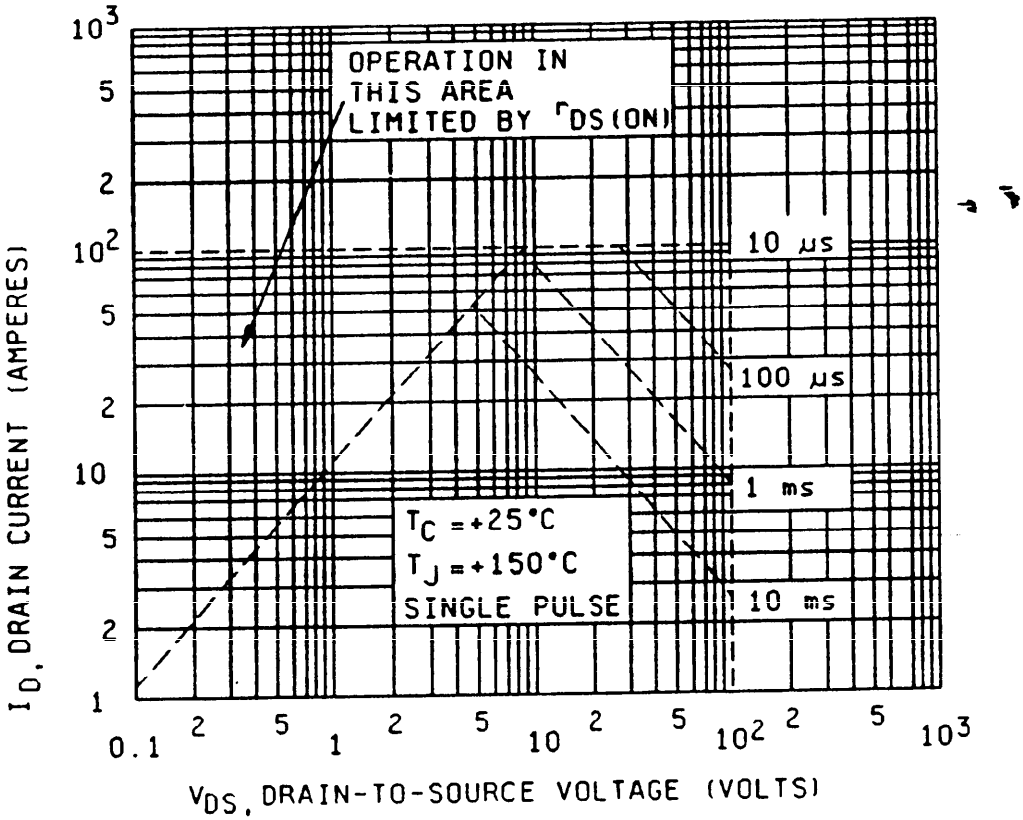


FIGURE 4. Thermal response curves.

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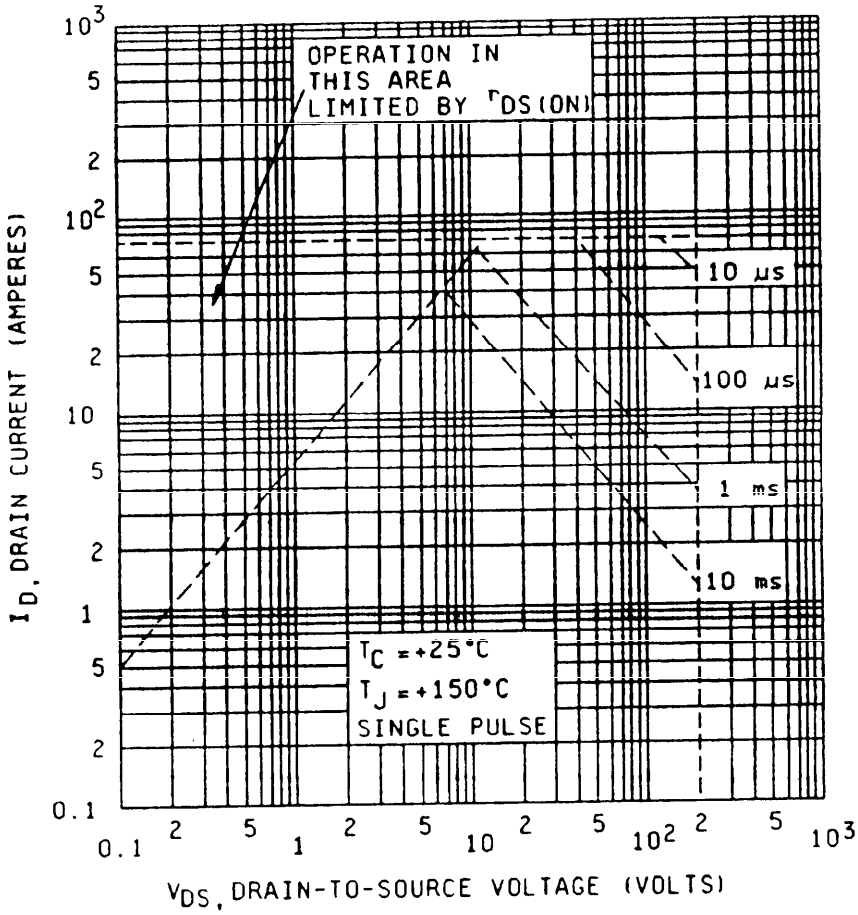


2N7236

NOTE: These curves refer to devices packaged in a TO-254AA package only. Derate curve by 20 percent for surface mount devices.

FIGURE 5. Safe operating area graph.

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2N7237

NOTE: These curves refer to devices packaged in a TO-254AA package only. Derate curve by 20 percent for surface mount devices.

FIGURE 5. Safe operating area graph - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish and formation as specified (see 3.3.1).

c. Type designation, product assurance level and for die acquisition, the manufacturer's letter identification should be specified.

6.3 Substitution information. Devices covered by this specification are one-way substitutable for the following manufacturer's Part or Identifying Number (PIN) listed below. This information in no way implies that manufacturers's PIN are suitable for the military PIN.

Military PIN	Manufacturer's CAGE code	Manufacturer's and User's PIN
2N7236	59993	IRFM 9140
2N7237	59993	IRFM 9240
2N7236U	59993	IRFN 9140
2N7237U	59993	IRFN 9240

6.4 Suppliers of die. The qualified die suppliers will be identified on the QPL (example, JANHCA7224).

JANC ordering information		
Military PIN	Manufacturer	
	59993	59993
2N7236	JANHCA2N7236	JANKCA2N7236
2N7237	JANHCA2N7237	JANKCA2N7237

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1618)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/595B	2. DOCUMENT DATE (YYMMDD) 950427

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT TRANSISTOR, P-CHANNEL, SILICON TYPES 2N7236, 2N7237, 2N7236U, 2N7237U, JANTX, JANTXV, JANS, JANHC AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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